Performed by: Vipul Jadhav

Assignment 10

Task: Build and Deploy WAR File on Slave Node Using Jenkins Master

Steps:

1. Launch one master instance and one slave instance

Jenkins-Master	i-06b4adef640f4efad	Running	t3.micro	3/3 checks passec	View alarms +	ap-south-1a
slave-1	i-0a30fb5820b39154a	Running	t3.micro	3/3 checks passec	View alarms +	ap-south-1a

2. Installed java, tomcat v10 & Jenkins in Jenkins master and maven & tomcat server in slave manually

3. Created environment variable for maven

```
# .bash_profile

# Get the aliases and functions
if [ -f ~/.bashrc ]; then
        . ~/.bashrc
fi

# User specific environment and startup programs
#
#
export MAVEN_HOME=/mnt/build-tools/apache-maven-3.9.13
export PATH=$MAVEN_HOME/bin:$PATH
```

4. In Jenkins tools section uncheck install maven automatically checkbox and add name of maven (added apache-maven-3.9.13) and add the location of maven home.

Maven installations

Maven installations ^ Edited

+ Add Maven

Maven

Name

apache-maven-3.9.13

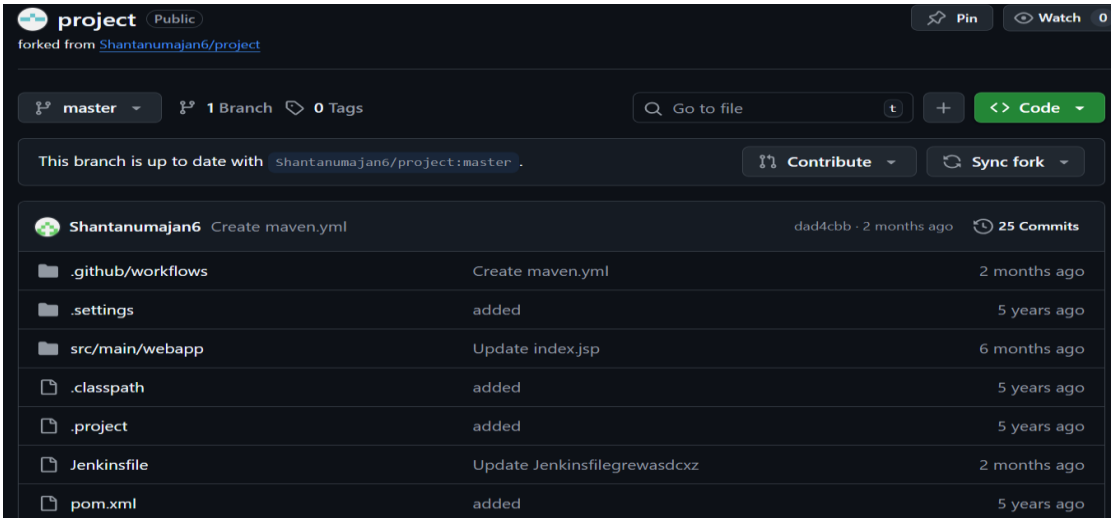
MAVEN_HOME

/mnt/build-tools/apache-maven-3.9.13

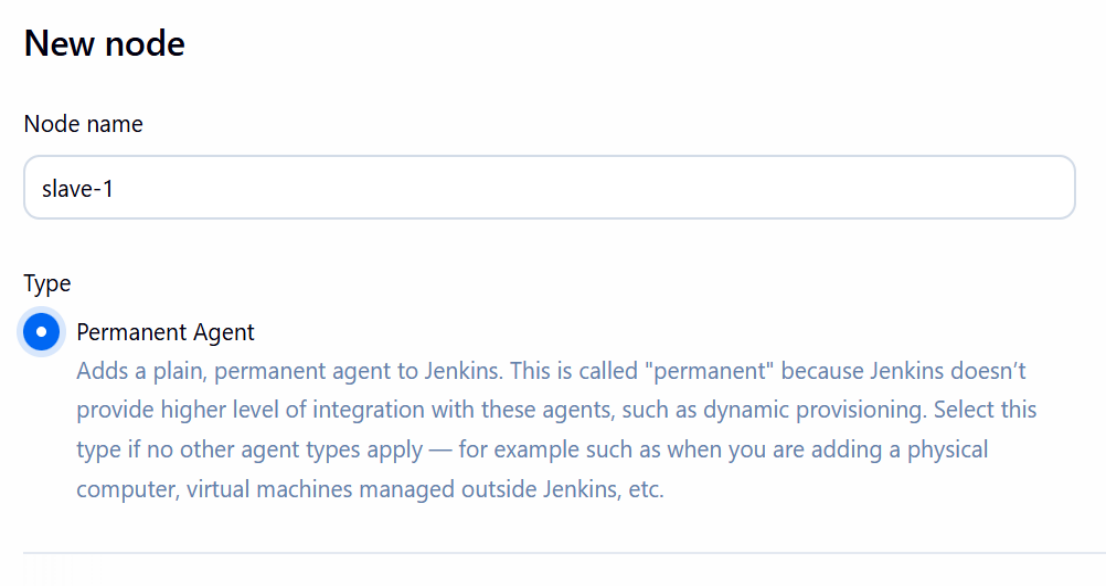
Install automatically ?

5. Created Jenkins workspace directory “/mnt/project” in master instance and “/mnt/Jenkins-slave” in slave instance

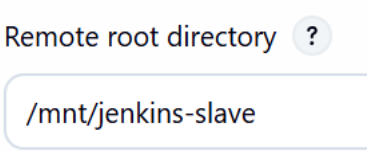
6. Github repository that we are going to use.



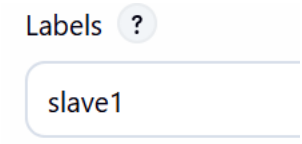
7. Creating slave-1 node and checking permanent agent option



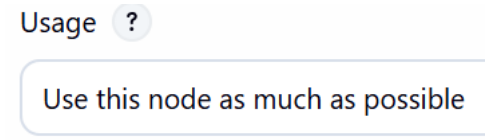
8. Created remote root directory in slave 1 and selected same remote root directory in “Remote root directory” section in slave configuration section



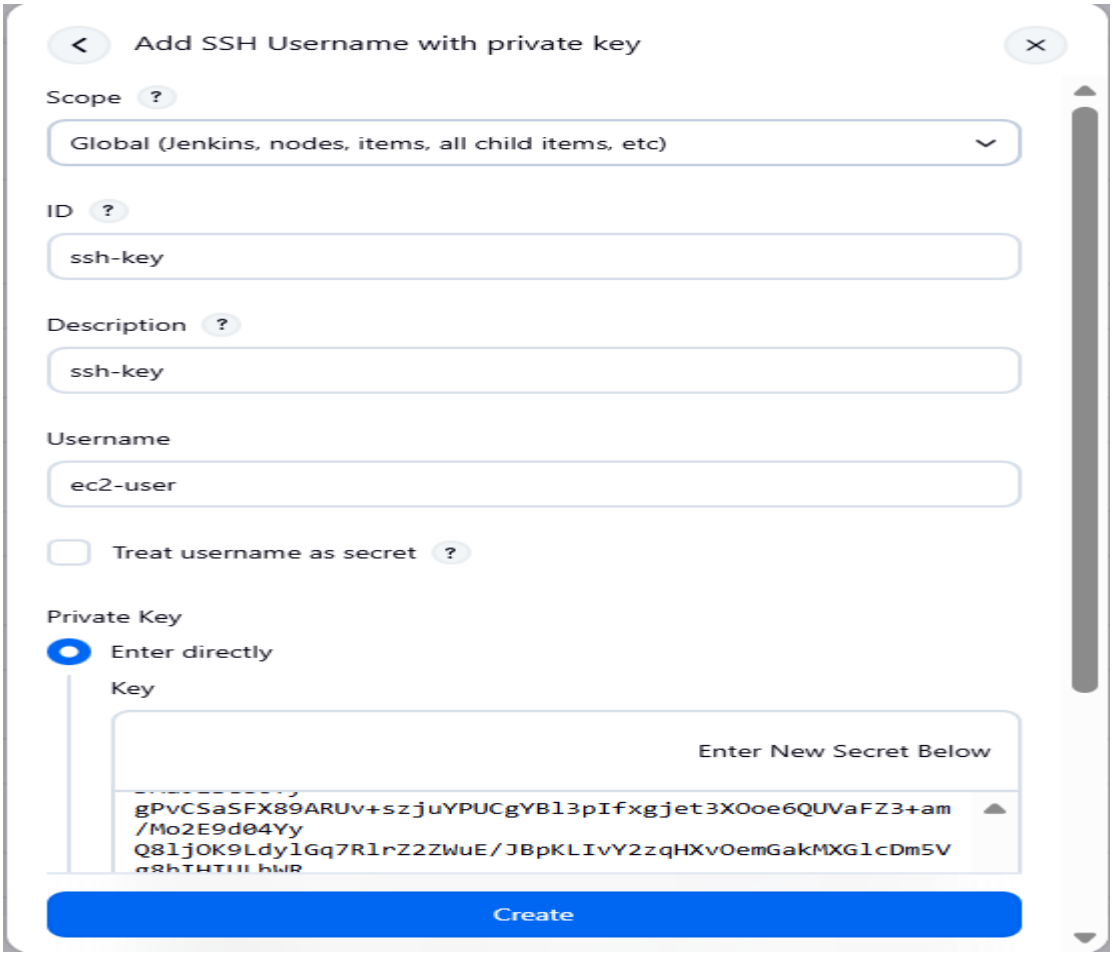
9. selected label slave 1 for slave 1 node



10. Selected usage as “Use this node as much as possible”

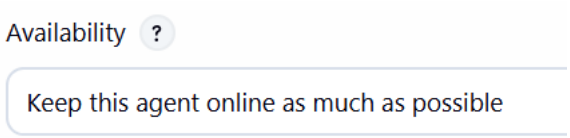


11. Created credentials as username with private key

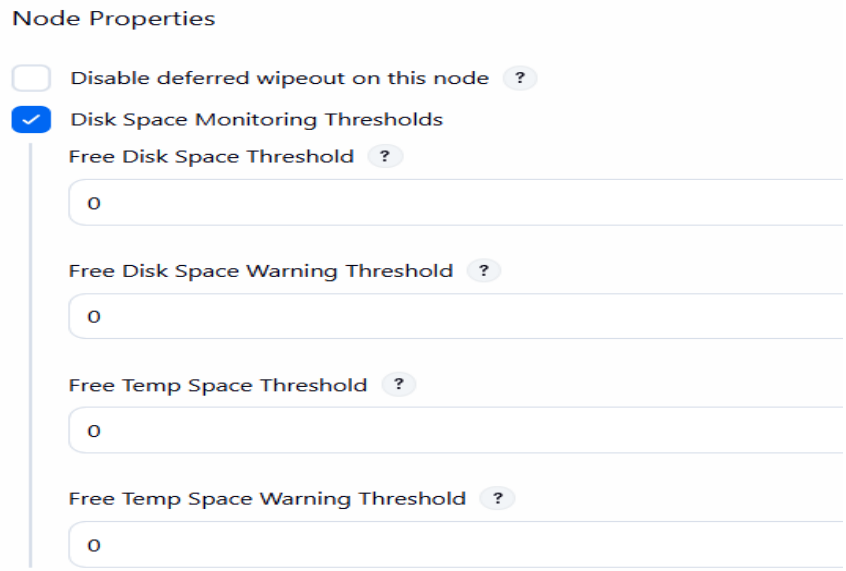


12. Selected launch method as “Launch agent via SSH” and provided private Ip and attached credentials and selected manually tested key verification strategy.

13. Selected availability as “Keep this agent online as much as possible”



14. Made all disk space monitoring thresholds as 0 in node properties and then clicked to apply and save



15. Now we have 1 slave node that we just created

S	Name ↓	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time	
	Built-In Node	Linux (amd64)	In sync	5.62 GiB	0 B	5.62 GiB	0ms	
	slave-1	Linux (amd64)	In sync	5.92 GiB	0 B	458.19 MiB	25ms	
	Data obtained	**56 min**	**56 min**	**56 min**	**56 min**	**56 min**	**56 min**	

16. Gave /mnt directory full permission

```
[root@ip-172-31-46-240 /]# chmod -R 777 mnt/
```

17. Creating freestyle job on master

New Item

Enter an item name

assignment-10

Select an item type

Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

18. Selected “Restrict where this project can run” as slave-1

Restrict where this project can be run ?

Label Expression ?

slave-1

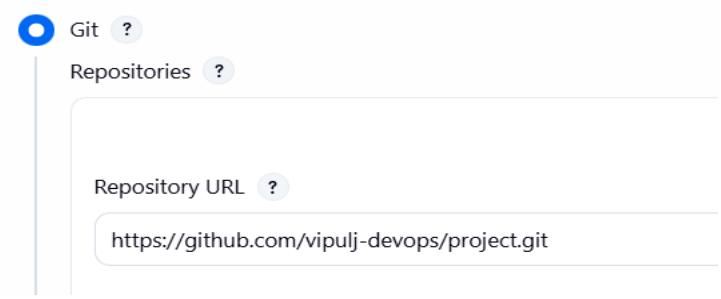
19. Added custom workspace

Use custom workspace ?

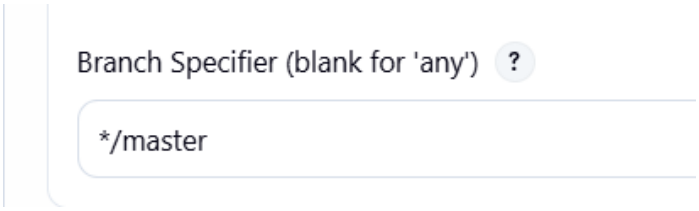
Directory

/mnt/jenkins-slave

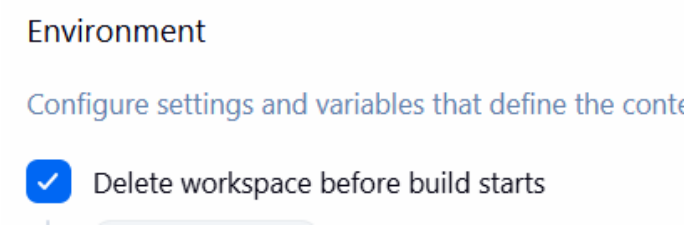
20. Put GitHub repository URL in GitHub URL section of Source Code Management



21. Select master branch in Branch Specifier

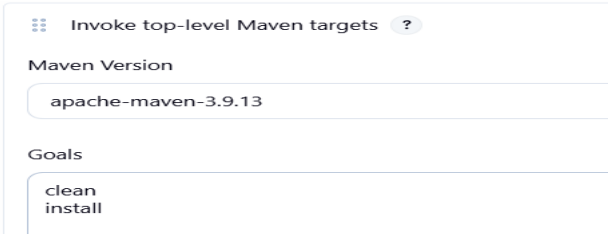


22. Check on 'Delete workspace before build starts' option. So, every time when we start new build the workspace will be cleared



23. Now in build step wrote first shell script to clean the maven repository

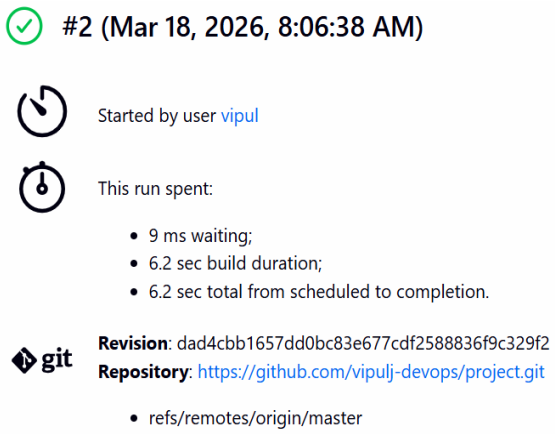
24. In build step selected Invoke top-level Maven targets configured Maven version and specified the goal in which mentioned two maven tasks (clean and install)



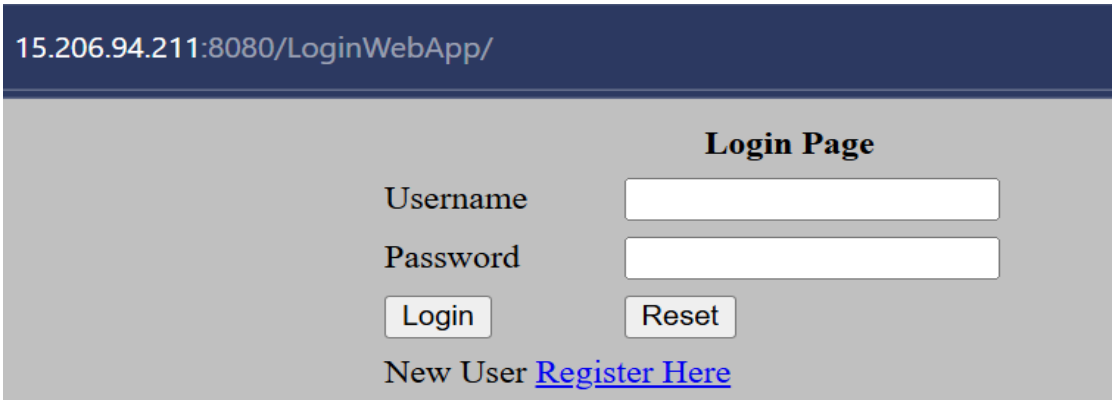
25. Wrote the below shell script in execute shell section and clicked on apply and save

```
#!/bin/bash

TOMCAT_PATH="/mnt/servers/apache-tomcat-10.1.52/webapps"
WAR_PATH="/mnt/jenkins-slave/target/LoginWebApp.war"

echo "Starting deployment on slave..."
# Step 1: Remove old deployment
rm -rf $TOMCAT_PATH/LoginWebApp*

# Step 2: Copy new WAR
cp $WAR_PATH $TOMCAT_PATH/

# Step 3: Restart Tomcat (optional but safer)
$TOMCAT_PATH/../bin/startup.sh

echo "Deployment completed successfully."
```

26. After clicking build now option the build ran successfully



27. Build ran on slave-1 instance

```
Started by user vipul
Running as SYSTEM
Building remotely on slave-1 in workspace /mnt/jenkins-slave
```

28. War file got hosted successfully on slave tomcat server